

EAD-01 Special (Before 24.1F batches) - 30th Jan 2025

Question 01

1. Write an object-oriented program to represent the following entity and perform the tasks below: (7 Marks)
 - a. Set values for two attributes of the entity created.
 - b. Display values of the attributes.
2. List out types of members which can be placed in an interface. (3 Marks)
3. The following codes are written to use class Phone by interface Device. But it generates errors. Please rewrite it to avoid errors. (7 Marks)

```
interface Device {  
    public void show()  
    {  
    }  
}  
  
class Phone extends Device {  
    {  
    }  
}
```

4. Write Java code to create a package called **business** to hold the class **Payment** which is visible to other packages. (3 Marks)

Question 02

1. Write the purpose of try and catch blocks in exception handling. (2 Marks)
2. List out three **unchecked exception** classes in Java. (3 Marks)
3. You are required to write necessary code to prevent crashing of the following code when it is executed: (5 Marks)

```
String test = null;  
int length = test.length();
```

4. Write a Java-based multi-threading application to display even and odd numbers from 1 to 50. Use two different threads for odd and even processes. (8 Marks)
5. List out two thread-blocking techniques. (2 Marks)

Question 03

1. Draw an image to illustrate how memory is created for a string. (3 Marks)
2. List out the tasks of the following string methods: (5 Marks)
 - a. indexOf()
 - b. charAt()
 - c. equals()
 - d. substring()
 - e. toUpperCase()
3. Why is the StringBuffer class considered thread-safe? (2 Marks)
4. What is the difference between Character Stream and Byte Stream? (5 Marks)
5. You are required to write a Java application to write 500 bytes into a file. (5 Marks)

Question 04

1. List five event classes provided by Java libraries. (5 Marks)
2. Briefly explain Java event handling mechanism. (5 Marks)
3. Write a Java event handling program to input the bill amount of a buyer and output the expected discounted bill amount. Assume you are given the following UI and coding space. Copy the UI to the answer sheet and name components properly. (10 Marks)

Question 05

1. You are given the database below and a table. Write codes to perform requested tasks.
 - a. Create a database connection to the database. (3 Marks)
 - b. Insert one record of a teacher. (3 Marks)
 - c. Select the code of a teacher when the number is given. (3 Marks)
 - Database: School
 - Table: Teacher
 - Fields:
 - Teacher Code: String
 - Teacher First Name: String
2. List down three Java design patterns. (3 Marks)
3. What is Generic in Java? Write an example applying generics. (8 Marks)